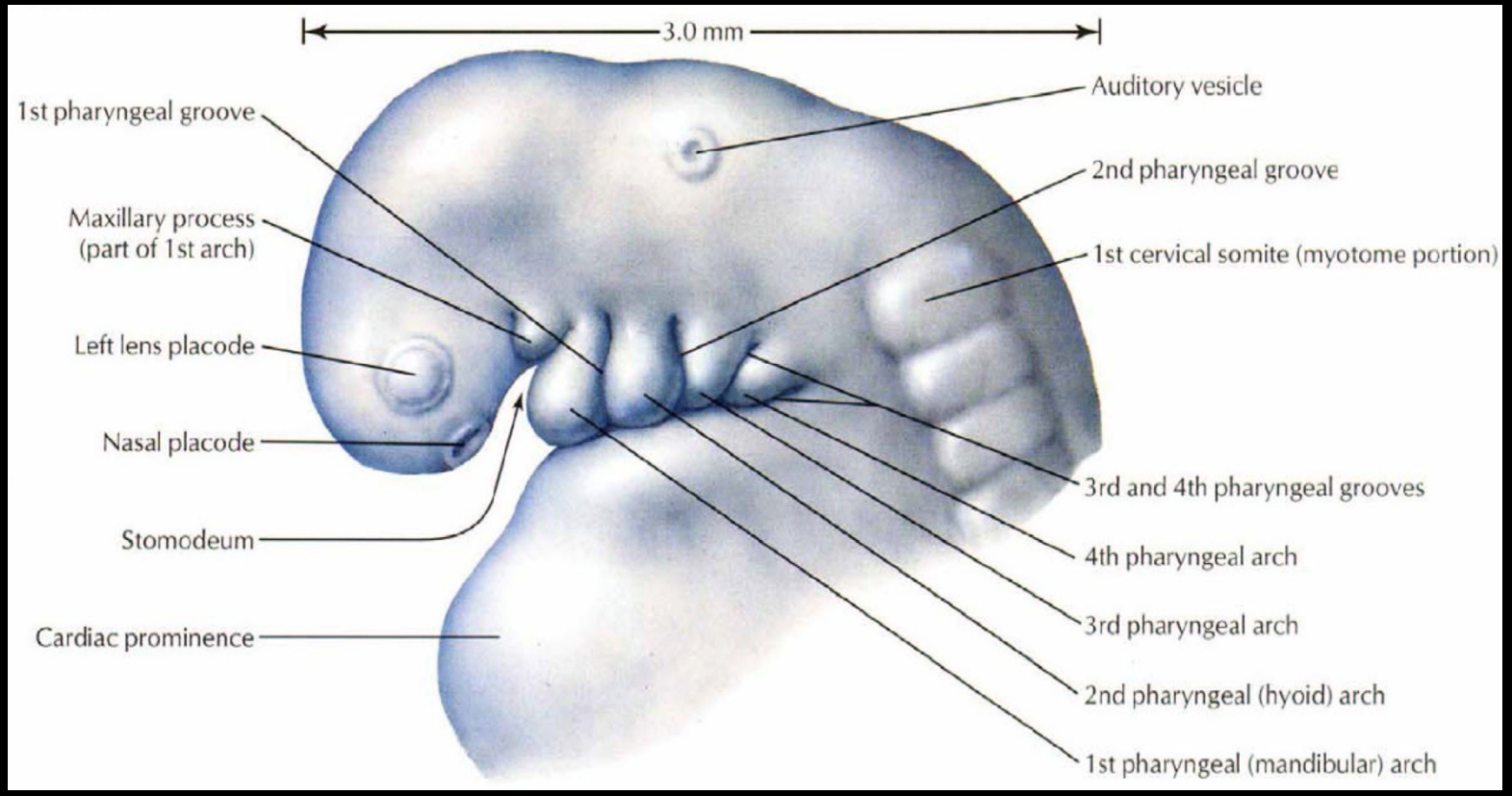

بسم الله الرحمن الرحيم

Development of the Face, Nose & Palate

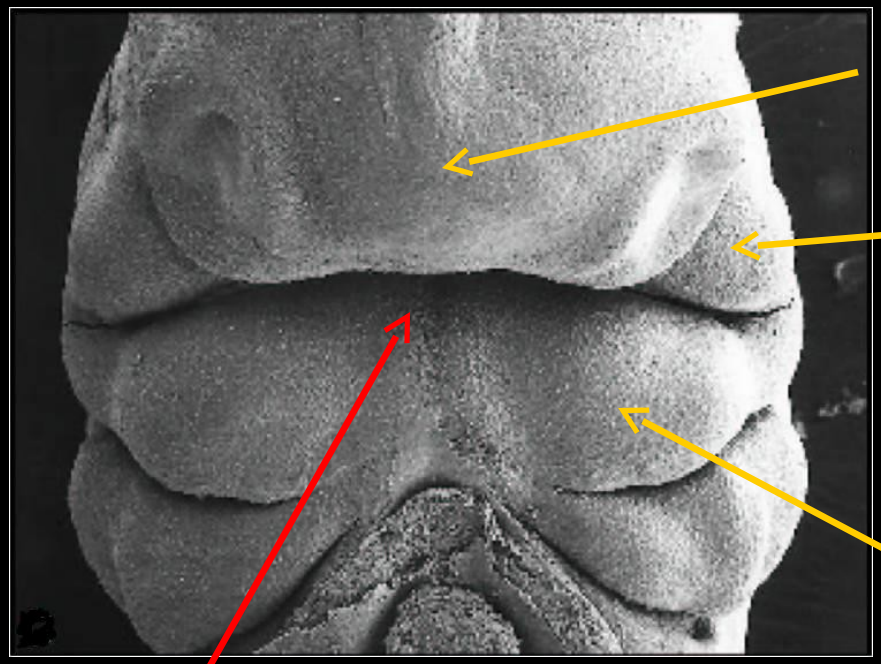
Development of the Face

- The development of the face occurs mainly between 5 – 8 weeks
- The lower jaw (mandible) is the first to form (4th week)
- The facial proportions develop during the fetal period (9th week to birth)
- During infancy & childhood, following the development of teeth and paranasal sinuses, the facial skeleton increases in size and contribute to the definitive shape of the face



Embryo at 4 - 5 weeks (Lateral view)

Early in the 4th week, five primordial swellings consisting primarily of neural crest-derived mesenchyme appear around the stomodeum and play an important role in the development of face



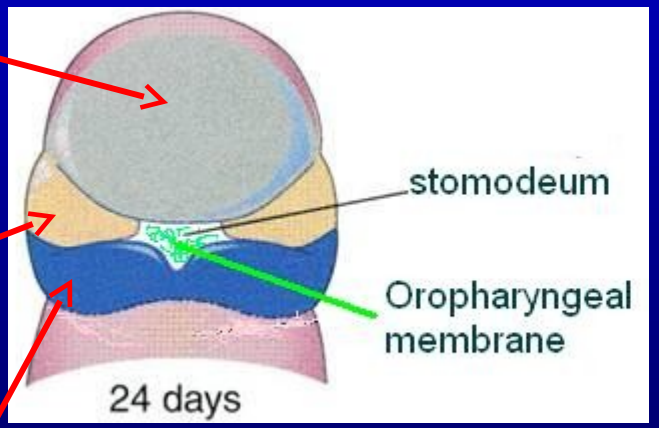
1 Frontonasal prominence

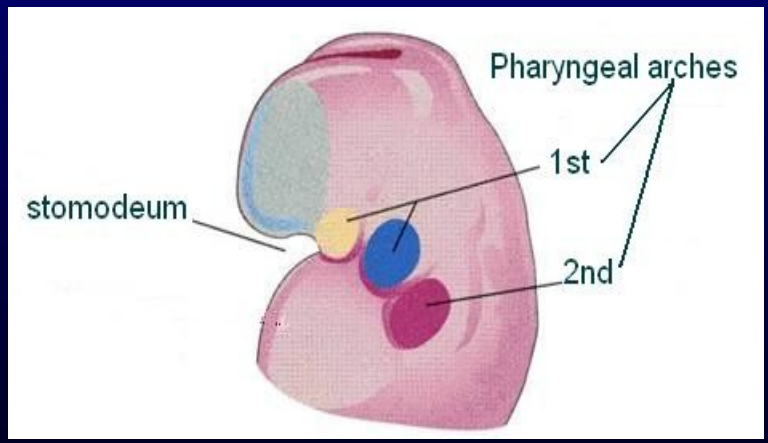
2 Maxillary prominences

2 Mandibular prominences

Stomodeum

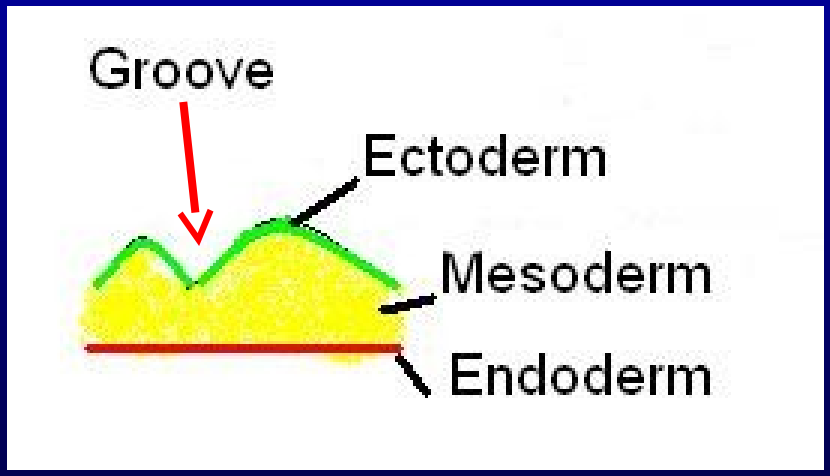
- The single frontonasal prominence ventral to the forebrain
- The paired maxillary prominences develop from the cranial part of first branchial arch
- The paired mandibular prominences develop from the caudal part of first branchial arch





Lateral view

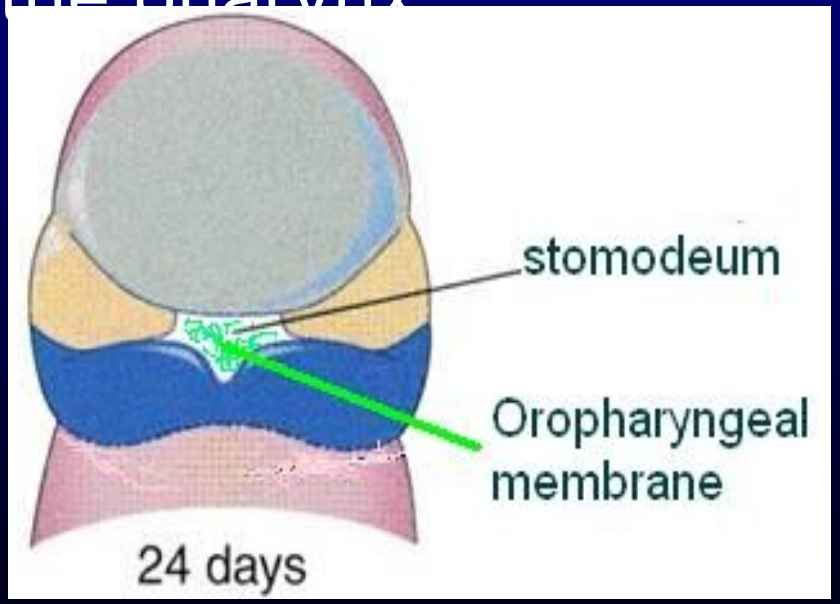
- The mesoderm of the five prominences is continuous with each other
- There is no internal division corresponding to the grooves demarcating the prominences externally



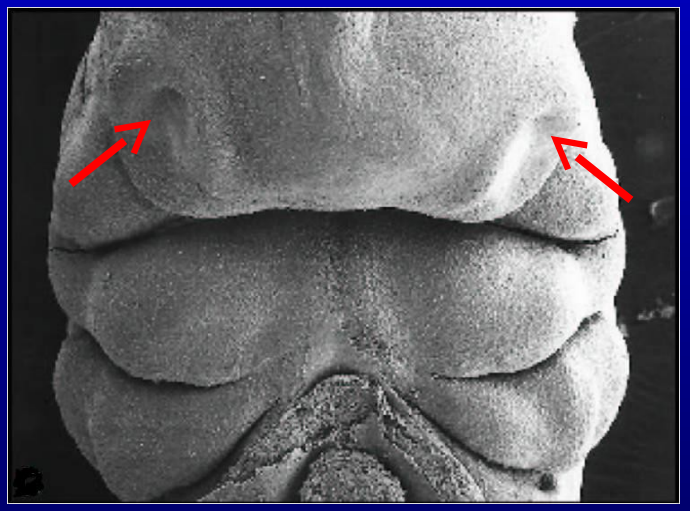
Stomodeum

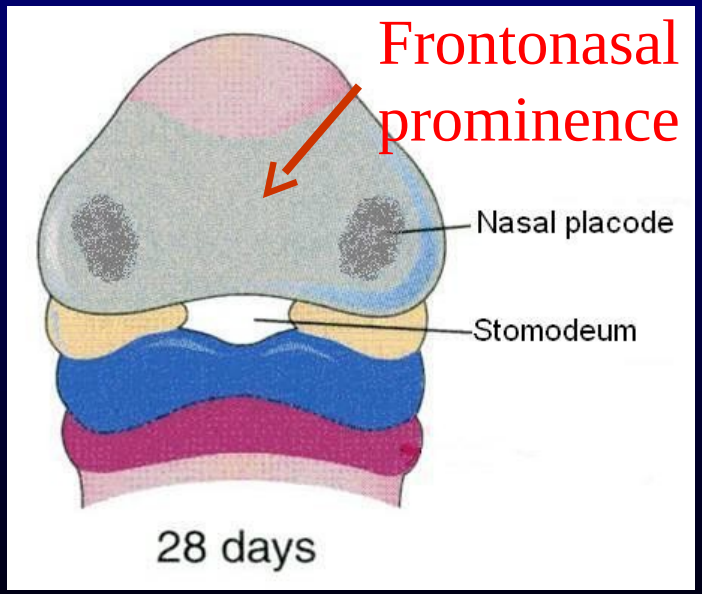
- An ectoderm lined depression
- Separated from the primitive pharynx by the buccopharyngeal (oropharyngeal) membrane
- The membrane later breaks down and stomodeum opens into the pharynx

Forms the vestibule of the oral cavity

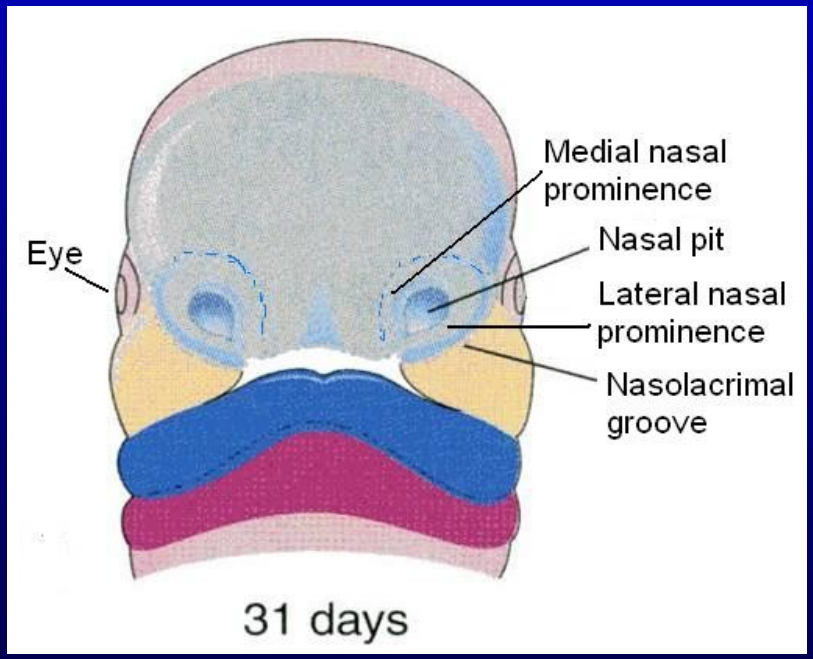


- By the end of 4th week, bilateral oval-shaped ectodermal thickenings called 'nasal placodes' appear on each side of the lower part of the frontonasal prominence
- Nasal placodes are primordia of the nose and nasal cavities.

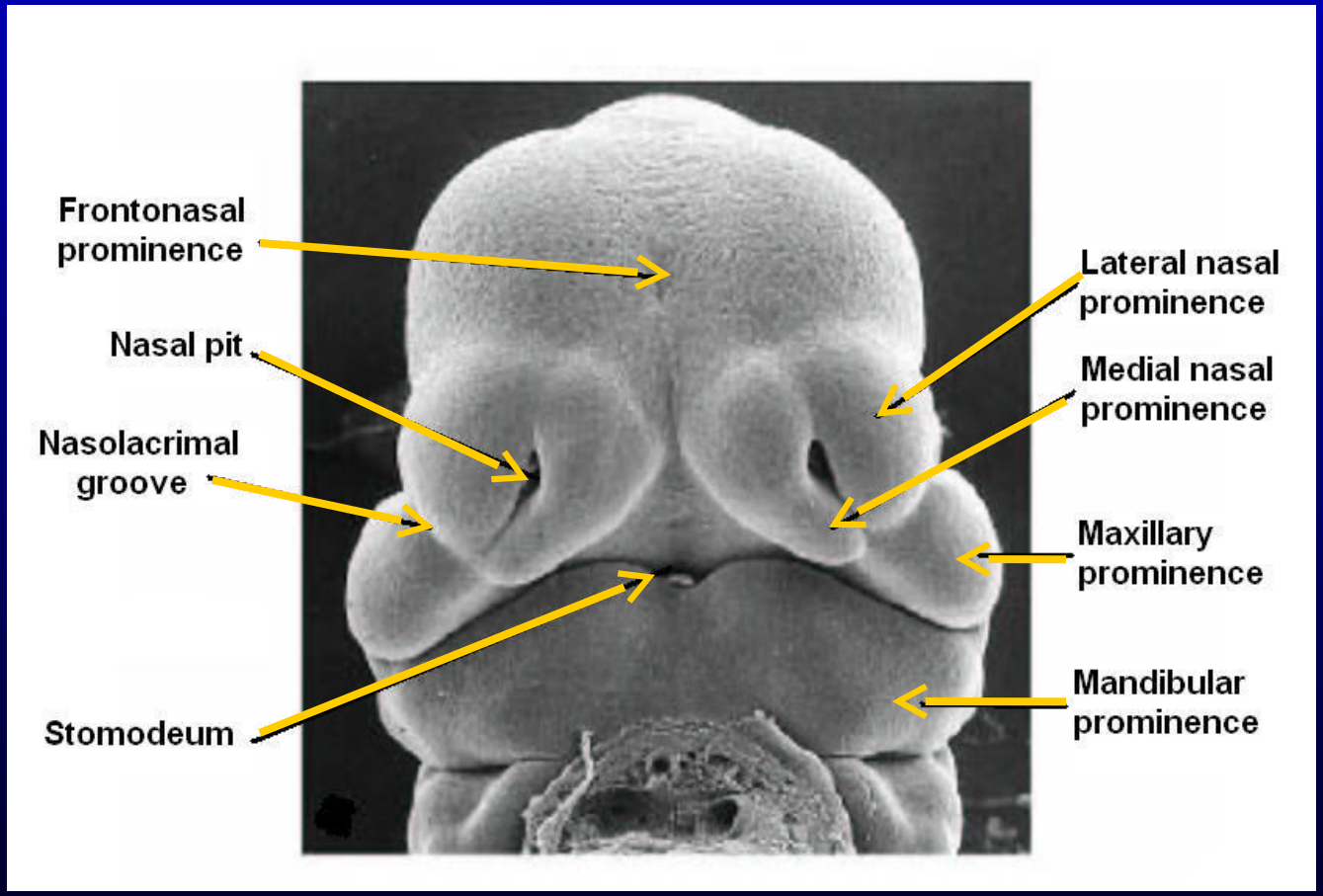




- Mesenchymal cells proliferate at the margin of the placodes and produce horse-shoe shaped swellings around these.
- The sides of these swellings are called ‘medial’ and ‘lateral’ nasal prominences
- The placodes now lie in the floor of a depression called ‘nasal pits’

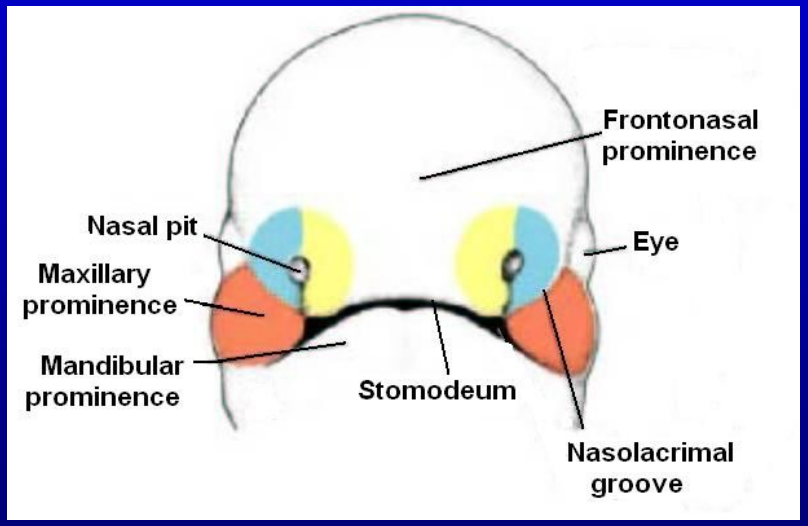


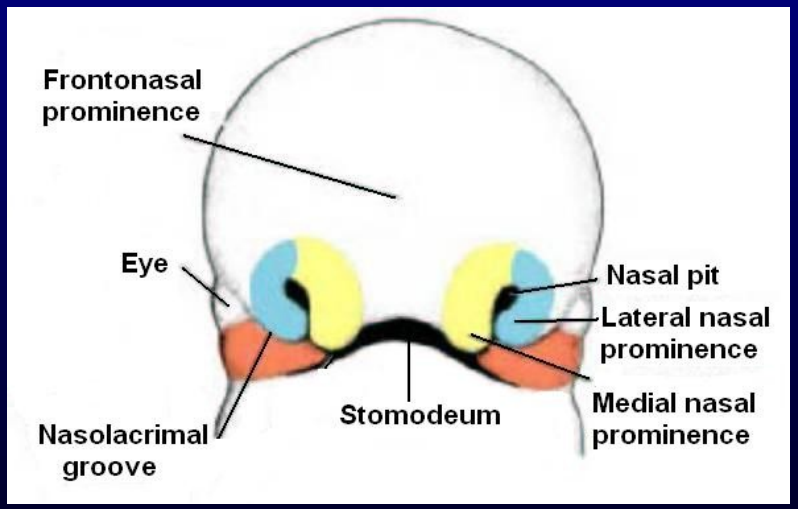
Each lateral nasal prominence is separated from the maxillary swelling by nasolacrimal groove



Mouse embryo: 6 weeks

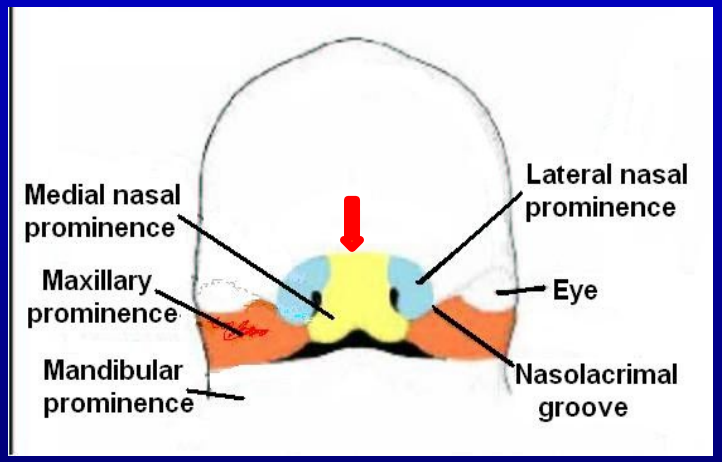
- The maxillary prominences continue to increase in size and:
- Laterally, merge with the mandibular prominences to form the cheek
- Medially, compress the medial nasal prominences toward the midline and finally fuses with these to form the upper lip.

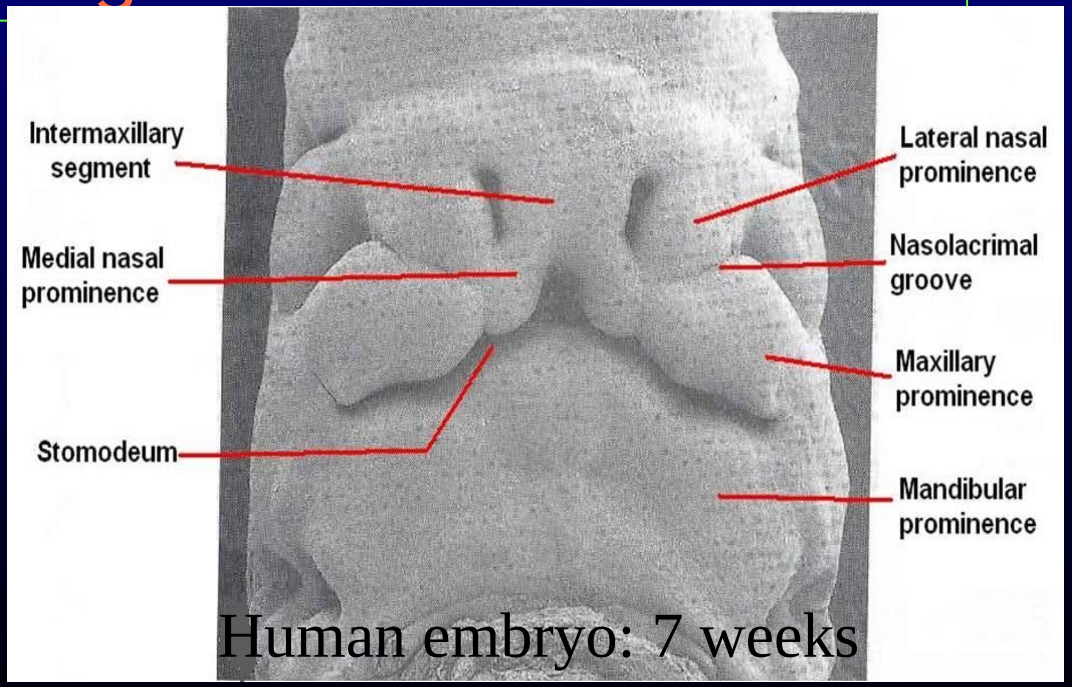




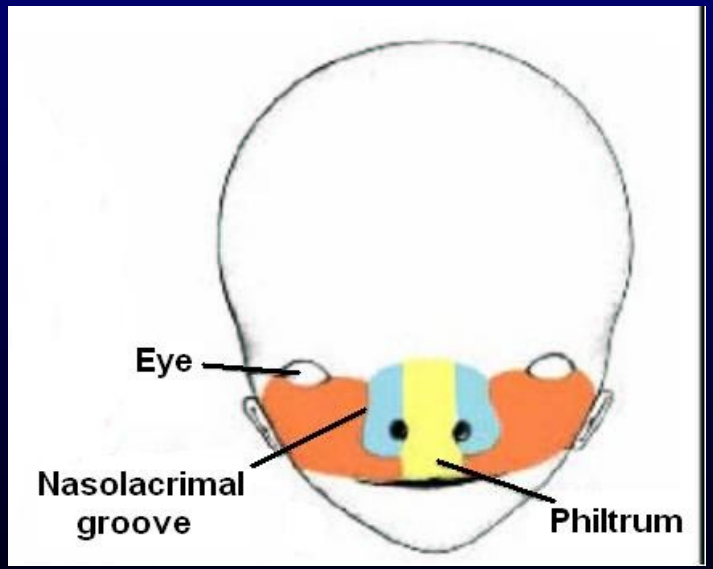
The upper lip is formed by the two medial nasal prominences & the two maxillary prominences

The medial nasal swellings enlarge, grow medially and merge with each other in the midline to form the intermaxillary segment





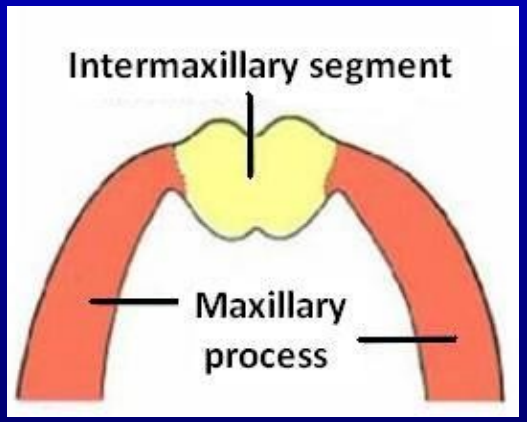
Human embryo: 7 weeks

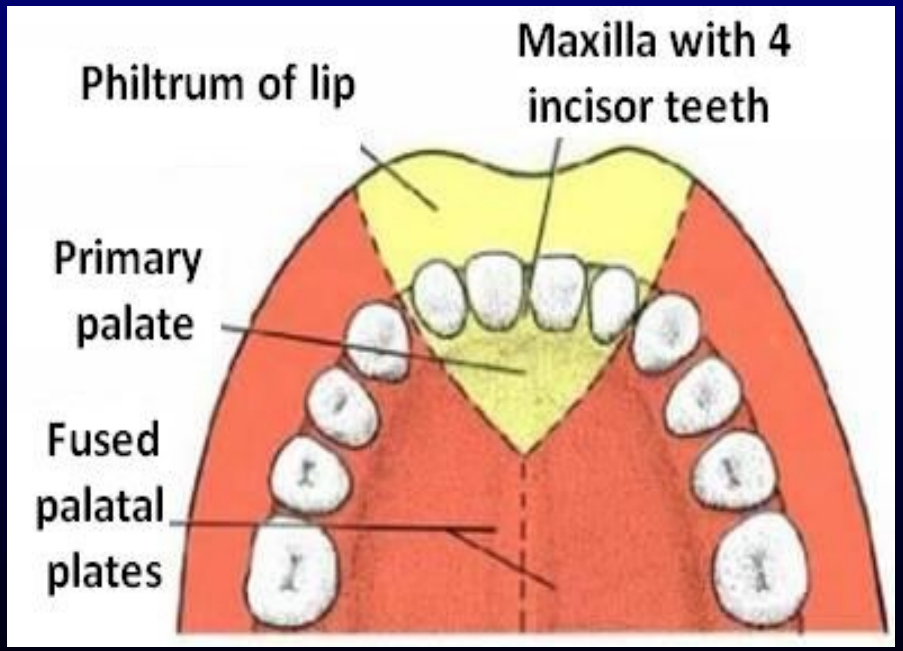


Intermaxillary Segment

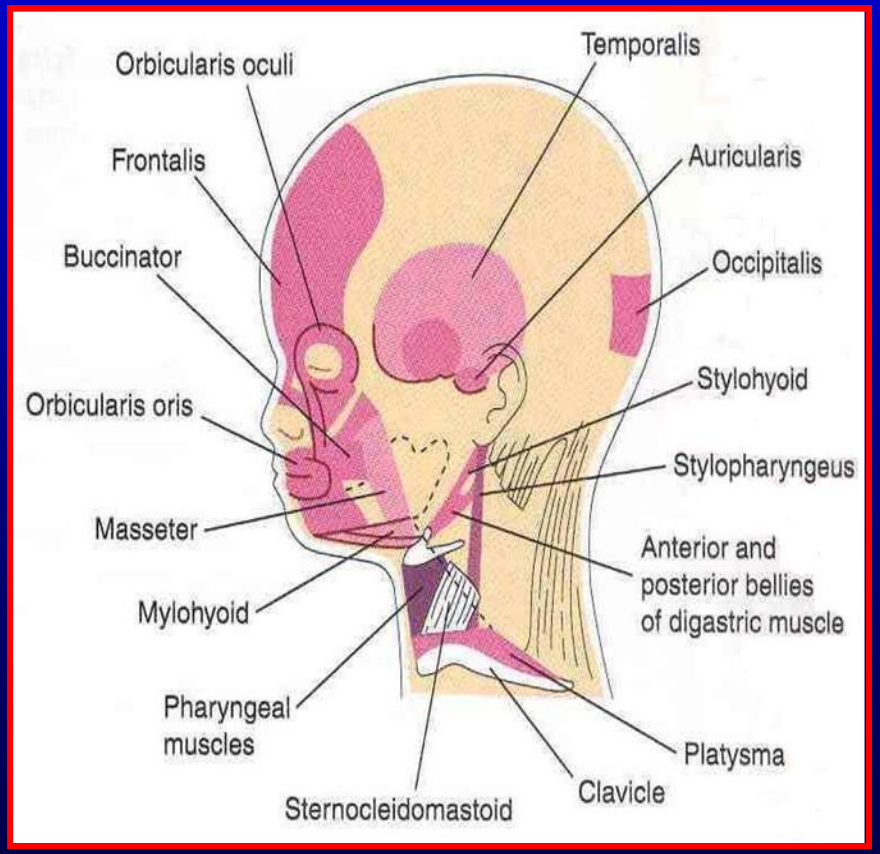
Gives rise to the:

- Philtrum of lip
- Premaxillary part of the maxilla, that bears the upper 4 incisors and the associated gums
- Primary palate (region of hard palate just posterior to the upper incisors)





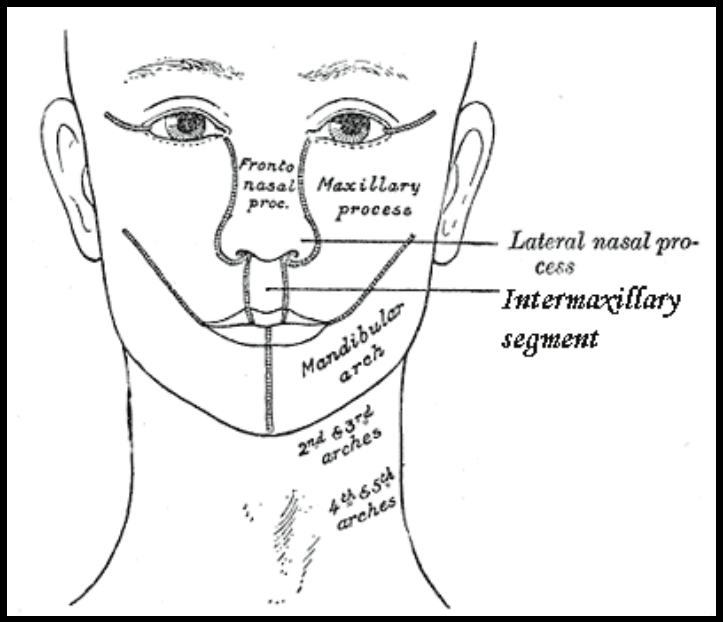
The mesenchyme from the 1st & 2nd pairs of pharyngeal arches invade the facial prominences and give rise to the muscles of mastication and muscles of facial expression respectively



Besides the fleshy derivatives, the facial prominences also give rise to bones of the facial skeleton

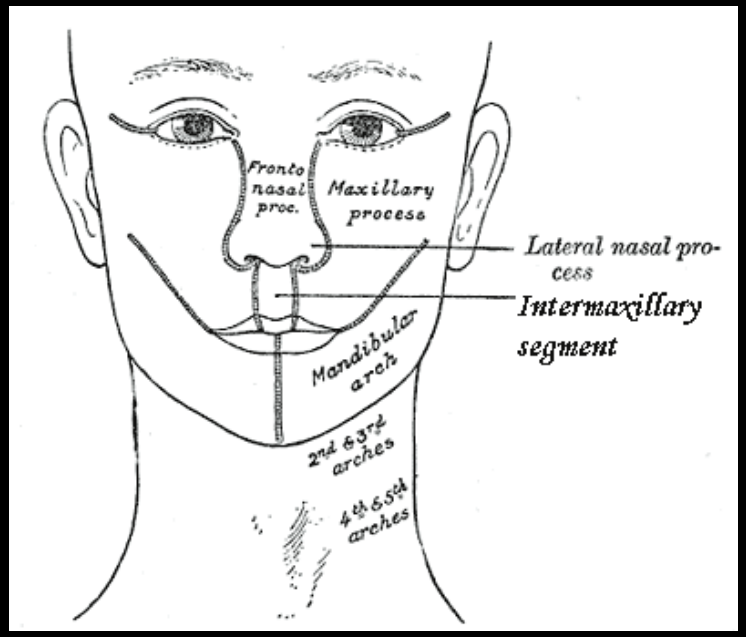
Derivatives of Facial Components

- The frontonasal prominence forms the:
  - Forehead and the bridge of the nose
  - Frontal and nasal bones



- The maxillary prominences form the:
  - Upper cheek regions and most of the upper lip
  - Maxilla, zygomatic bone, secondary palate

- The mandibular prominences fuse and form the:
  - Chin, lower lip, and lower cheek regions
  - Mandible

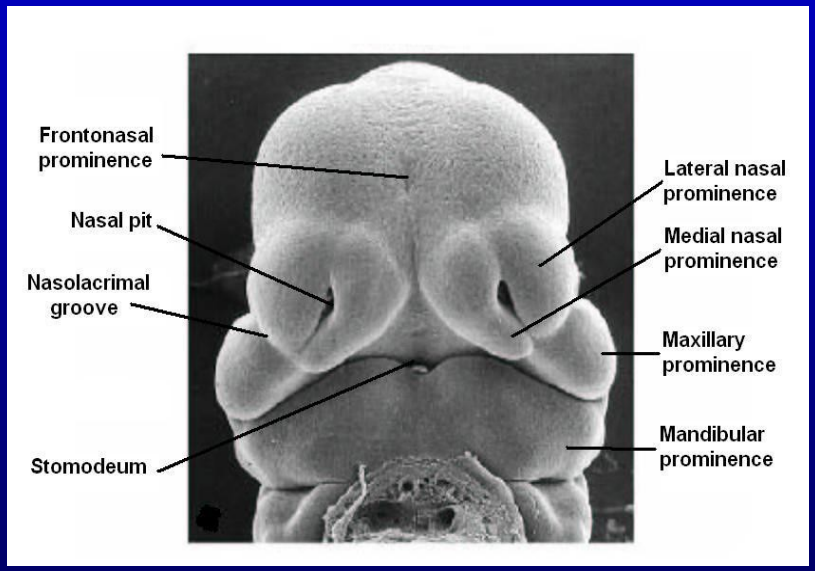


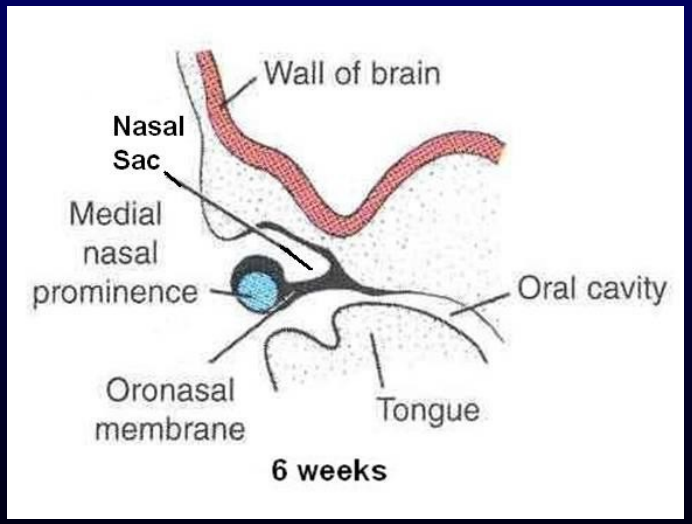
- The lateral nasal prominences form the alae of the nose

- The medial nasal prominences fuse and form the intermaxillary segment

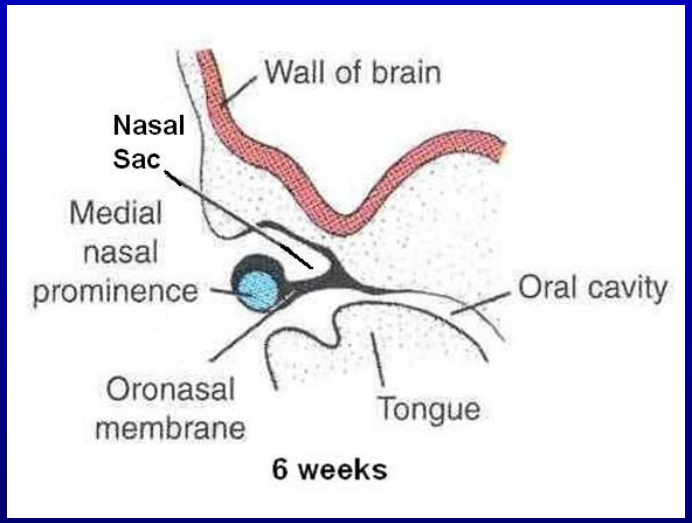
Development of the Nasal Cavity & Paranasal Sinuses

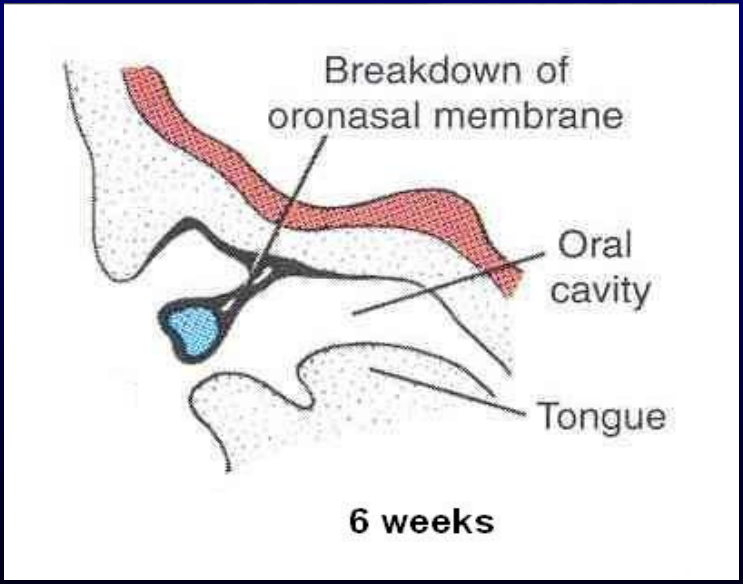
- With the formation of the medial and lateral nasal prominences, the nasal placodes lie in the floor of depressions called the nasal pits
- By the end of 6th week, nasal pits deepen and form nasal sacs
- Each nasal sac grows dorsocaudally, ventral to the developing brain



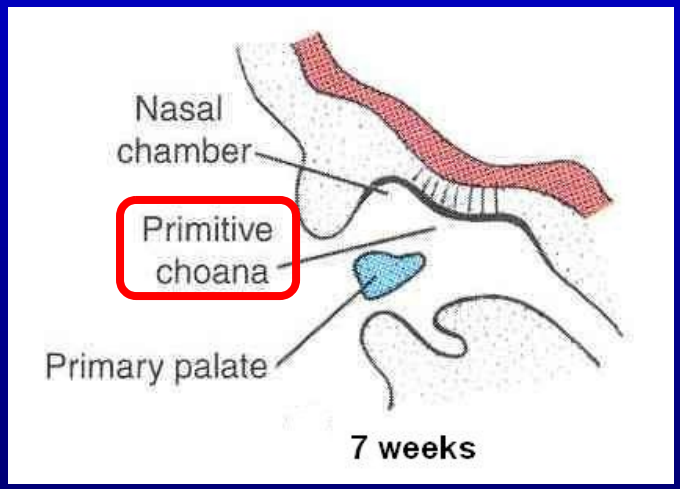


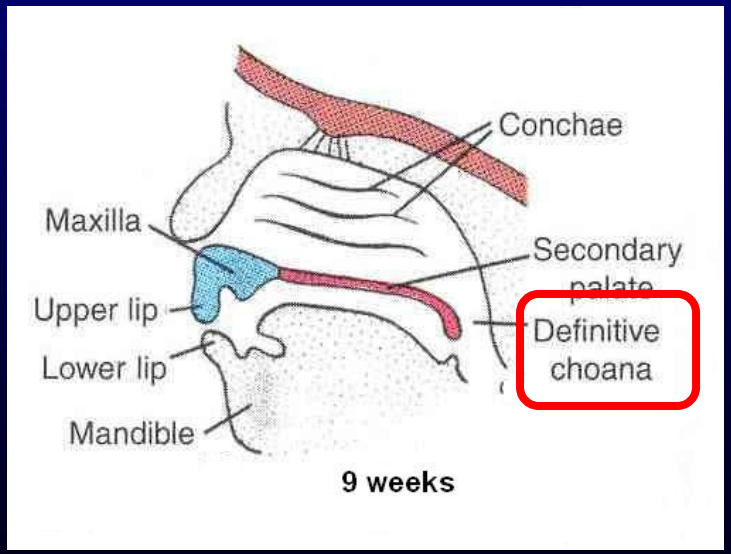
- Initially the nasal sacs are separated from the oral cavity by oronasal membrane.
- The oronasal membrane ruptures by the 7th week, communicating the primitive nasal cavities with the oral cavity



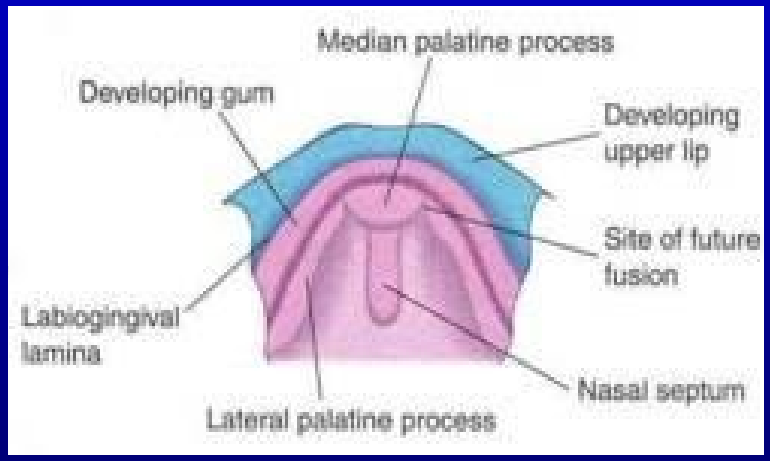


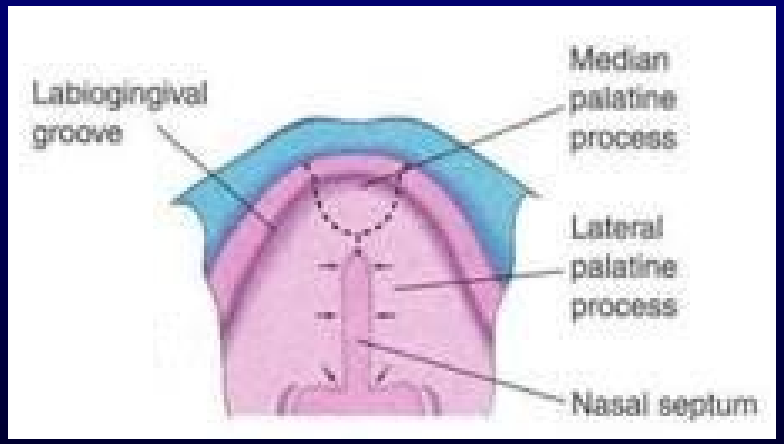
- These communications are called the primitive choanae and are located posterior to the primary palate
- After the development of the secondary palate, the choanae change their position and become located at the junction of nasal cavity and the pharynx



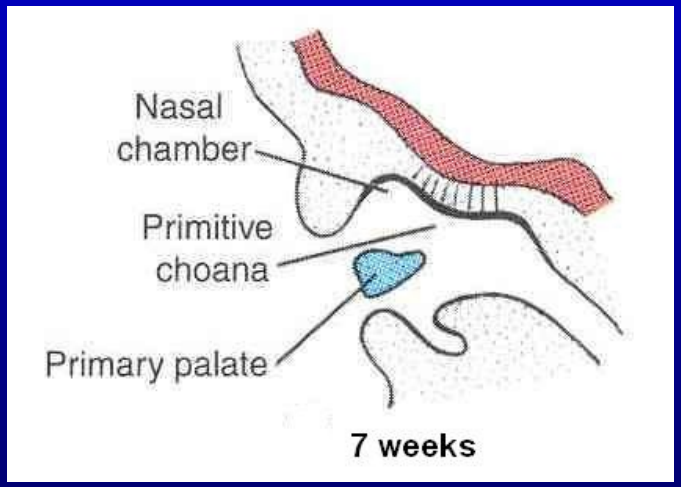


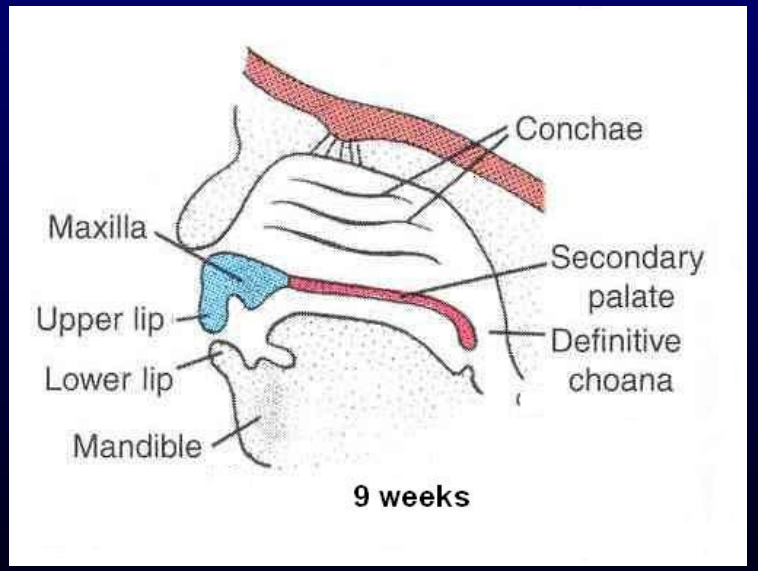
- The nasal septum develops as a downgrowth from the internal parts of merged medial nasal prominences
- Fuses with the palatine process in 9-12 weeks, superior to the hard palate primordium



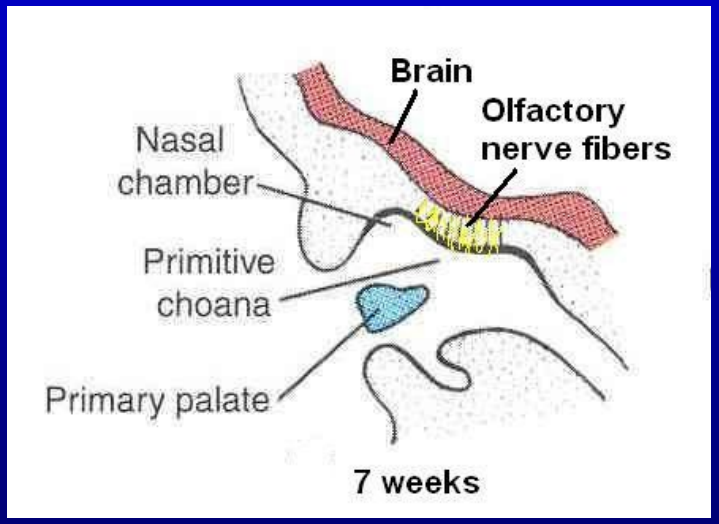


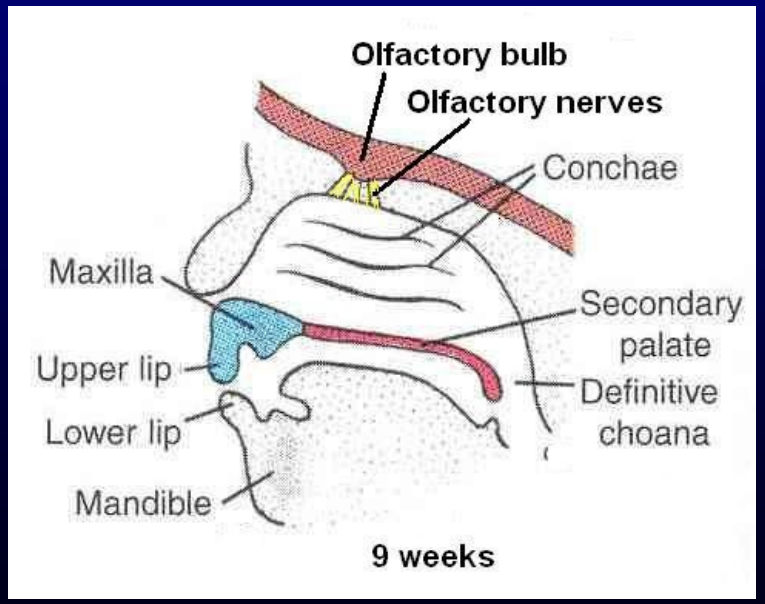
- The superior, middle and inferior conchae develop on the lateral wall of each nasal cavity
- The ectodermal epithelium in the roof of each nasal cavity becomes specialized as the olfactory epithelium



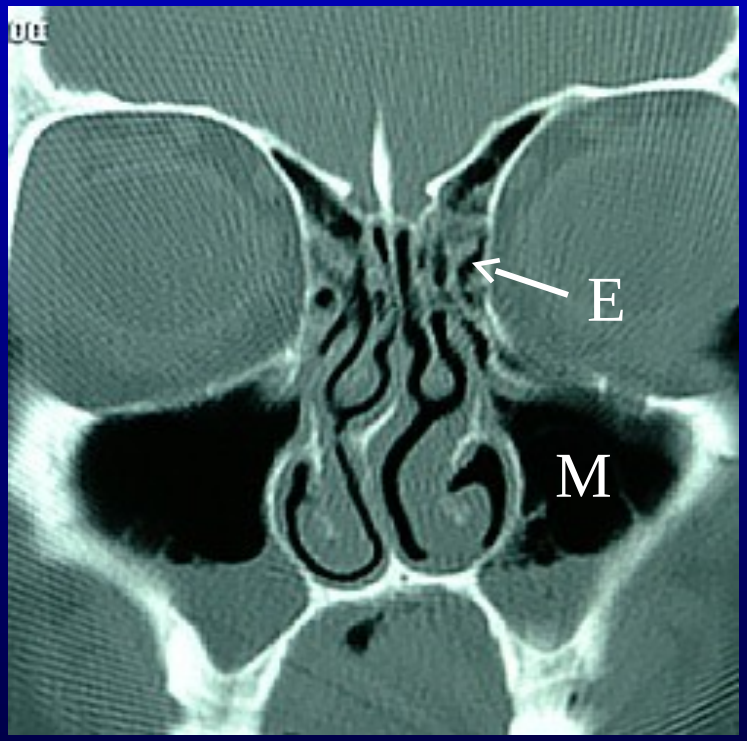


- The olfactory cells of the olfactory epithelium give origin to olfactory nerve fibers that grow into the olfactory bulb





- The paranasal sinuses develop as diverticulae of the walls of the nasal cavity
- Maxillary sinuses and few anterior & posterior ethmoidal air cells develop in fetal life
- Frontal and sphenoidal sinuses develop after birth



From a 3 months old fetus, showing ethmoid & maxillary sinuses

Nasolacrimal duct

- Develops from a rod-like thickening of the ectoderm in the floor of the nasolacrimal groove
- This solid cord of cells separates from the surface ectoderm and lies in the underlying mesenchyme
- The cord gets canalized to form the nasolacrimal duct
- The cranial end of the duct expands to form the lacrimal sac
- The caudal end opens into the inferior meatus of the nasal cavity
- The duct is usually becomes completely patent only after birth
- Failure of complete canalization of the duct leads to atresia of the duct (seen in about 6% of newborn infants)

Development of Palate (Palatogenesis)

Palatogenesis

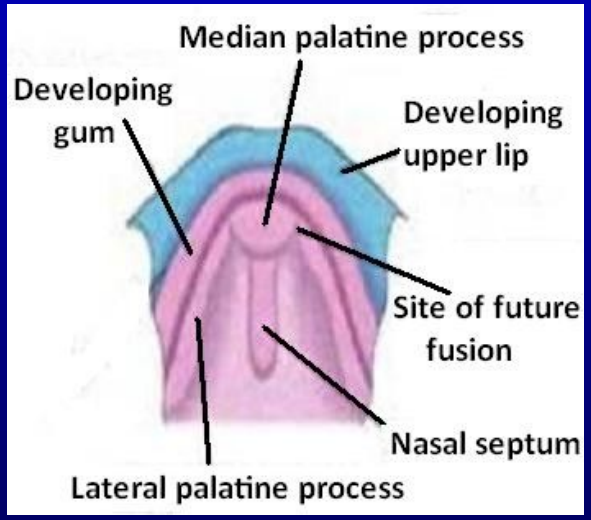
- Begins at the end of the 5th week
- Gets completed by the end of the 12th week
- The most critical period for the development of palate is from the end of 6th week to the beginning of 9th week

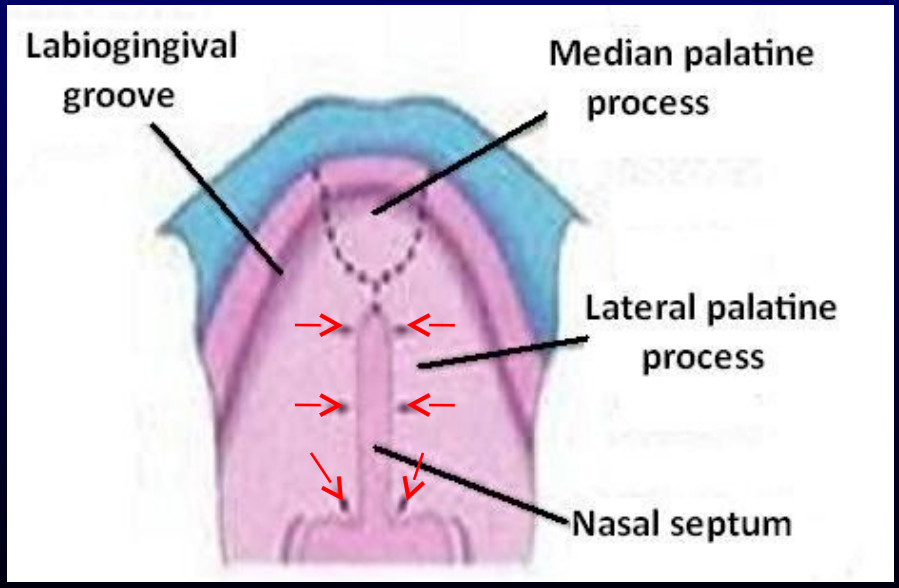
The palate develops from two primordia:

- The Primary palate
- The Secondary palate

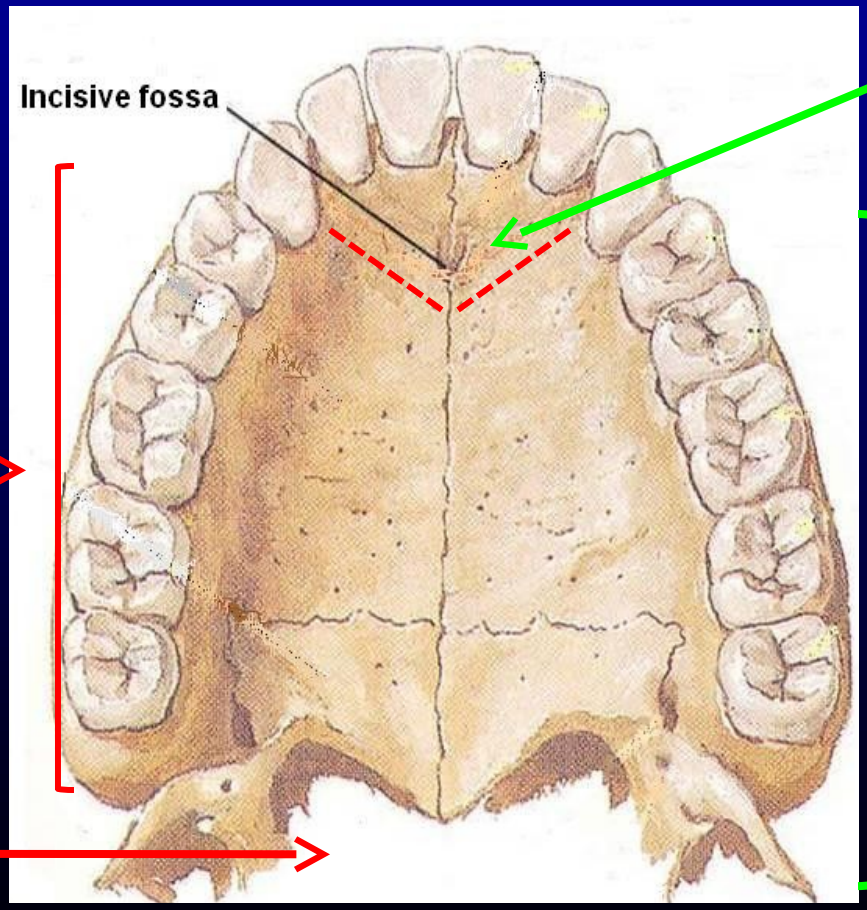
The Primary Palate

- Begins to develop:
  - Early in the 6th week
  - From the deep part of the intermaxillary segment, as median palatine process
- Lies behind the premaxillary part of the maxilla
- Fuses with the developing secondary palate



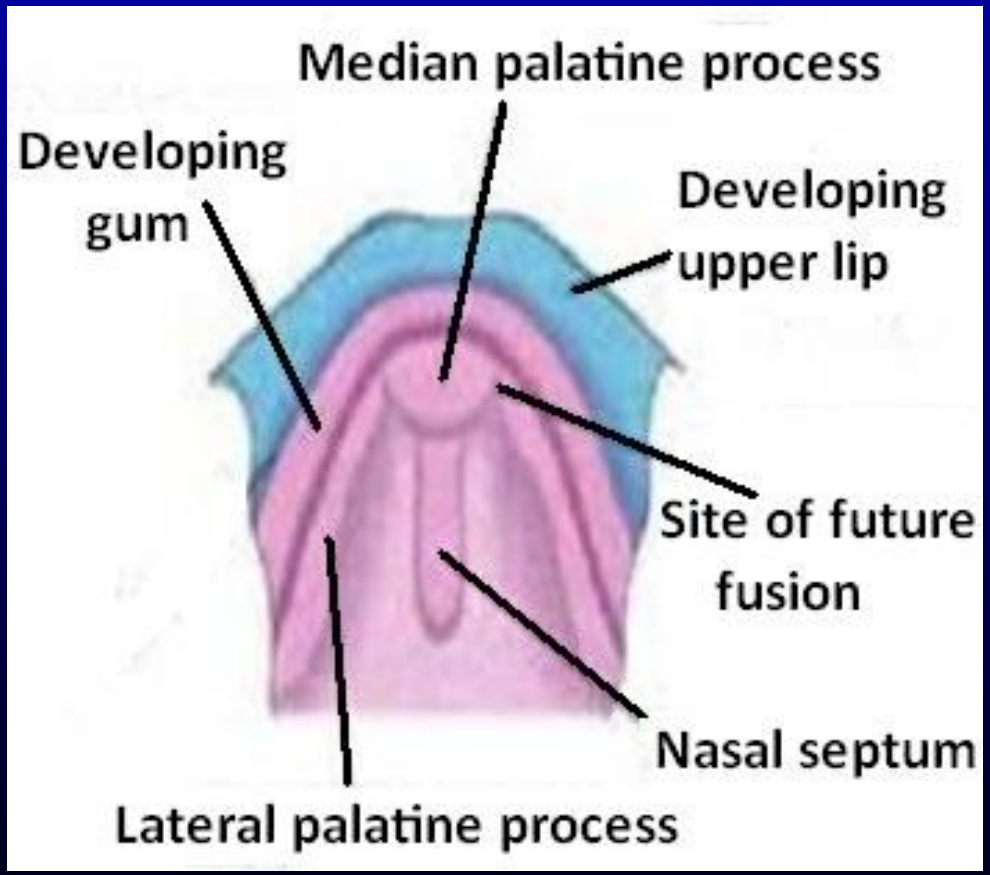


The primary palate represents only a small part lying anterior to the incisive fossa, of the adult hard palate

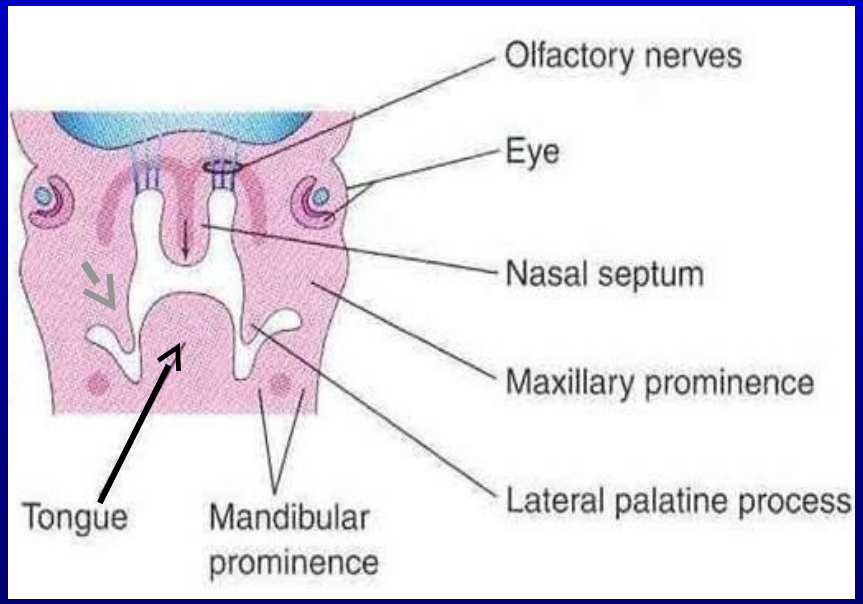


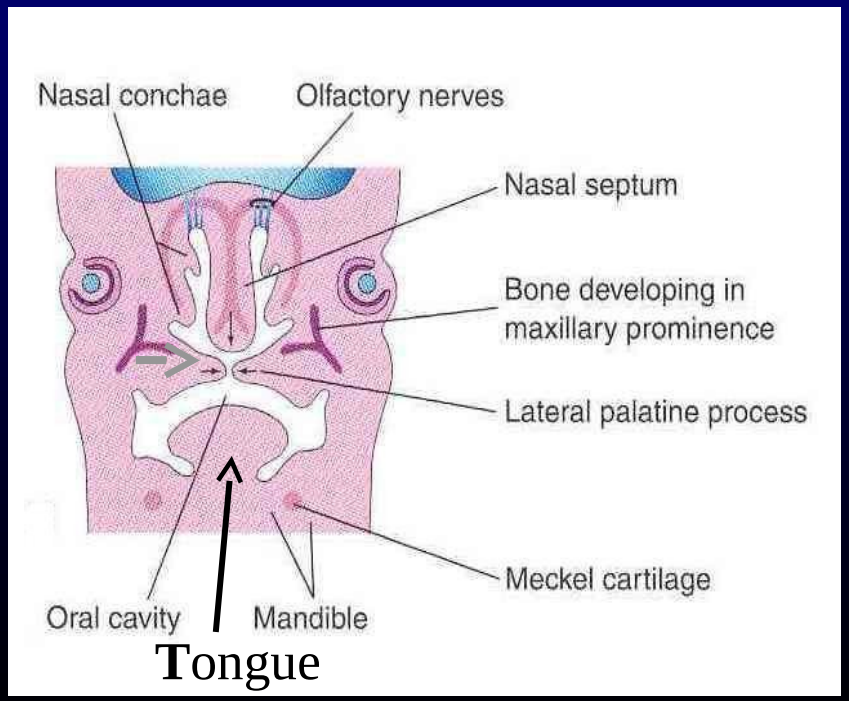
The Secondary Palate

- Is the primordia of hard and soft palate posterior to the incisive fossa
- Begins to develop:
  - Early in the 6th week
  - From the internal aspect of the maxillary processes, as lateral palatine process

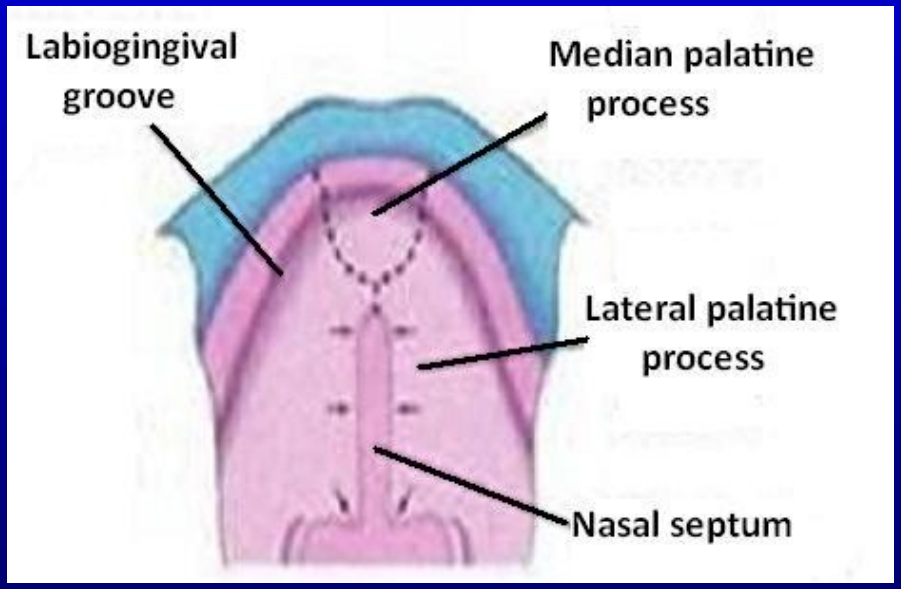


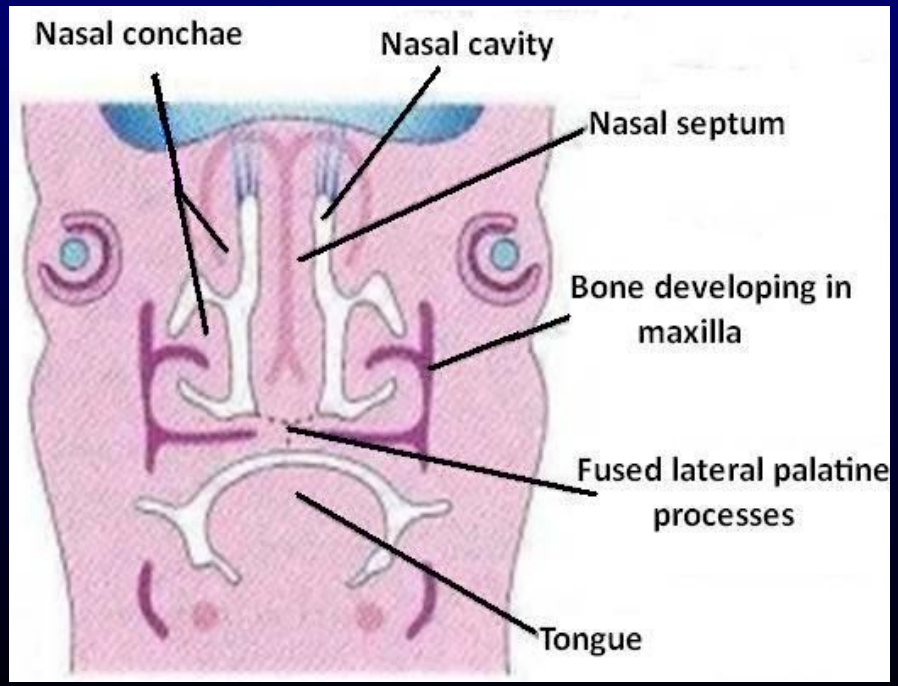
- In the beginning, the lateral palatine processes project inferomedially on each side of the tongue
- With the development of the jaws, the tongue moves inferiorly.
- During 7th & 8th weeks, the lateral palatine processes elongate and ascend to a horizontal position above the tongue



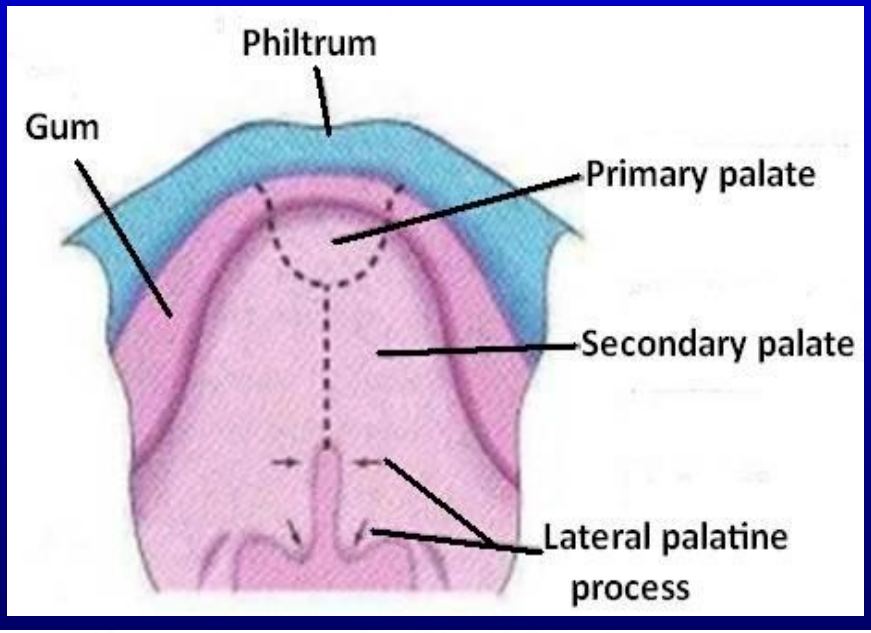


- Gradually the lateral palatine processes:
  - Grow medially and fuse in the median plane
  - Also fuse with the:
    - Posterior part of the primary palate
      &
    - The nasal septum

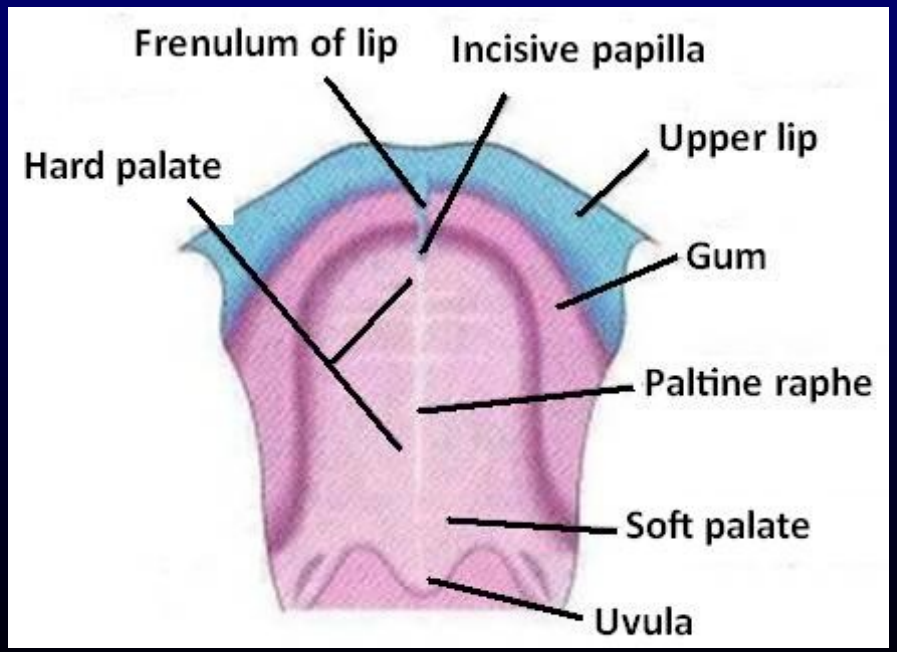




- Fusion with the nasal septum begins anteriorly during 9th week, extends posteriorly and is completed by 12th week

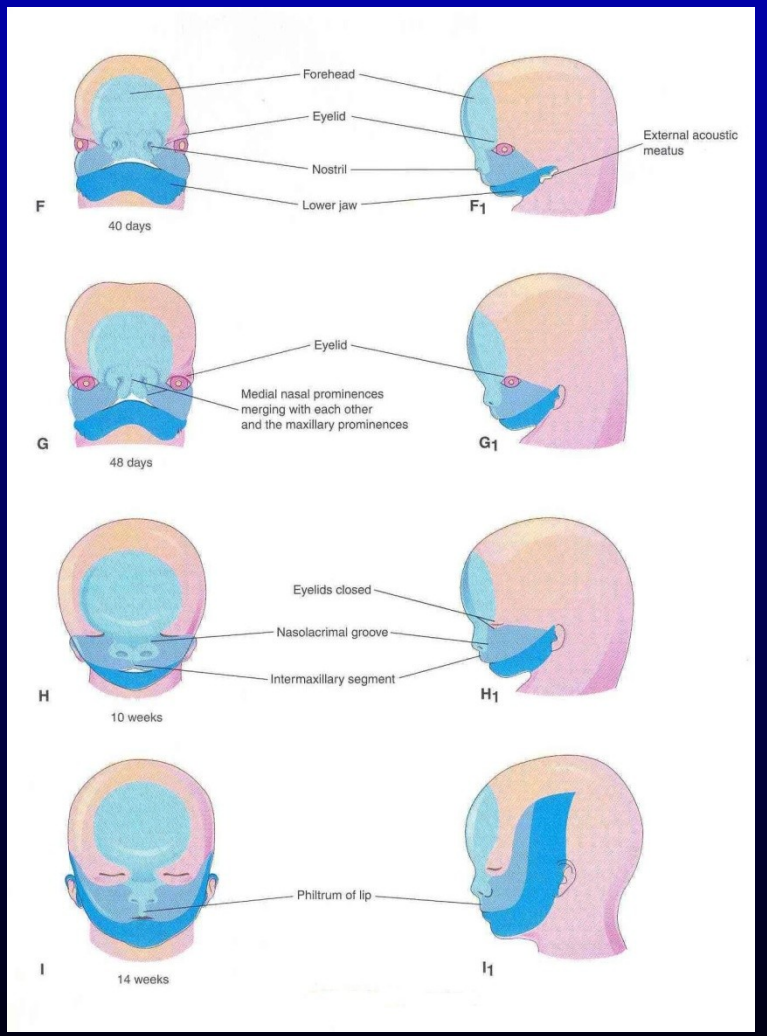


Bone develops in the anterior part to form the hard palate. The posterior part develops as muscular soft palate



Changes in Face during Fetal period

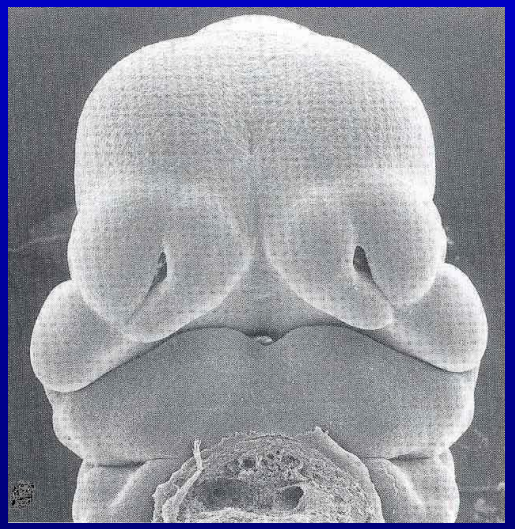
- Mainly result from changes in the proportion & relative positioning of facial structures
- In early fetal period the nose is flat and mandible underdeveloped. They attain their characteristic form during fetal period
- The enlargement of brain results in the formation of a prominent forehead
- Eyes initially appear on each side of frontonasal prominence move medially
- Ears first appear on lower portion of lower jaw, grow in upper direction to the level of the eyes
-



Anomalies related to Face, Nose & Palate

Facial clefts

Failure of the embryonic facial prominences to fuse properly



- May be unilateral or bilateral
- May involve:
  - Lips only: Cleft lip
  - Palate only: Cleft palate
  - Lip & palate: Cleft lip & palate
  - Region of nasolacrimal groove: Facial clefts

Lead to difficulty in breathing feeding sucking swallowing & speech

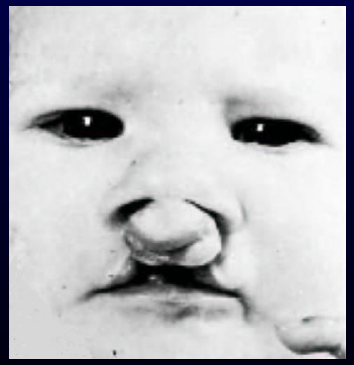
- **Median cleft lip:** results from failure of the medial nasal prominences to merge and form the intermaxillary segments
- **Unilateral cleft lip:** result from failure of the maxillary prominence to merge with the medial nasal prominence on the affected side
- **Bilateral cleft lip:** results due to failure of maxillary prominences to meet and unite with the medial nasal prominences on both sides



Median Cleft lip



Unilateral cleft lip

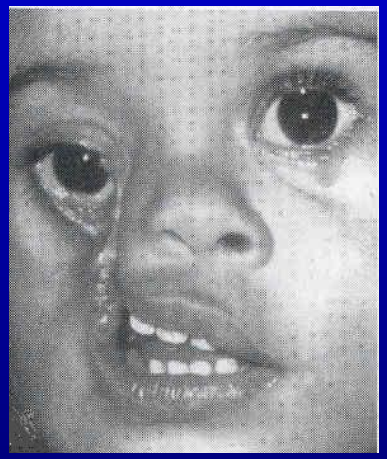


Bilateral cleft lip

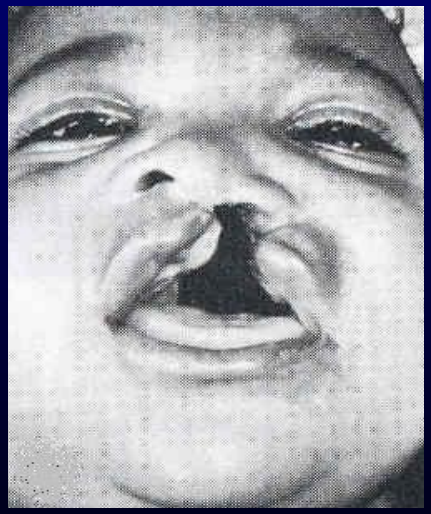
2. **Oblique facial cleft**: results from failure of the maxillary prominence to fuse with the lateral nasal prominence
3. **Cleft palate** leaves the nasal and oral cavities connected & results in nursing problem for the new born

May be:

- Anterior/posterior to incisive foramen
- Unilateral/bilateral
- Isolated/associated with cleft lips



Oblique facial cleft



Cleft lip, cleft jaw & cleft palate

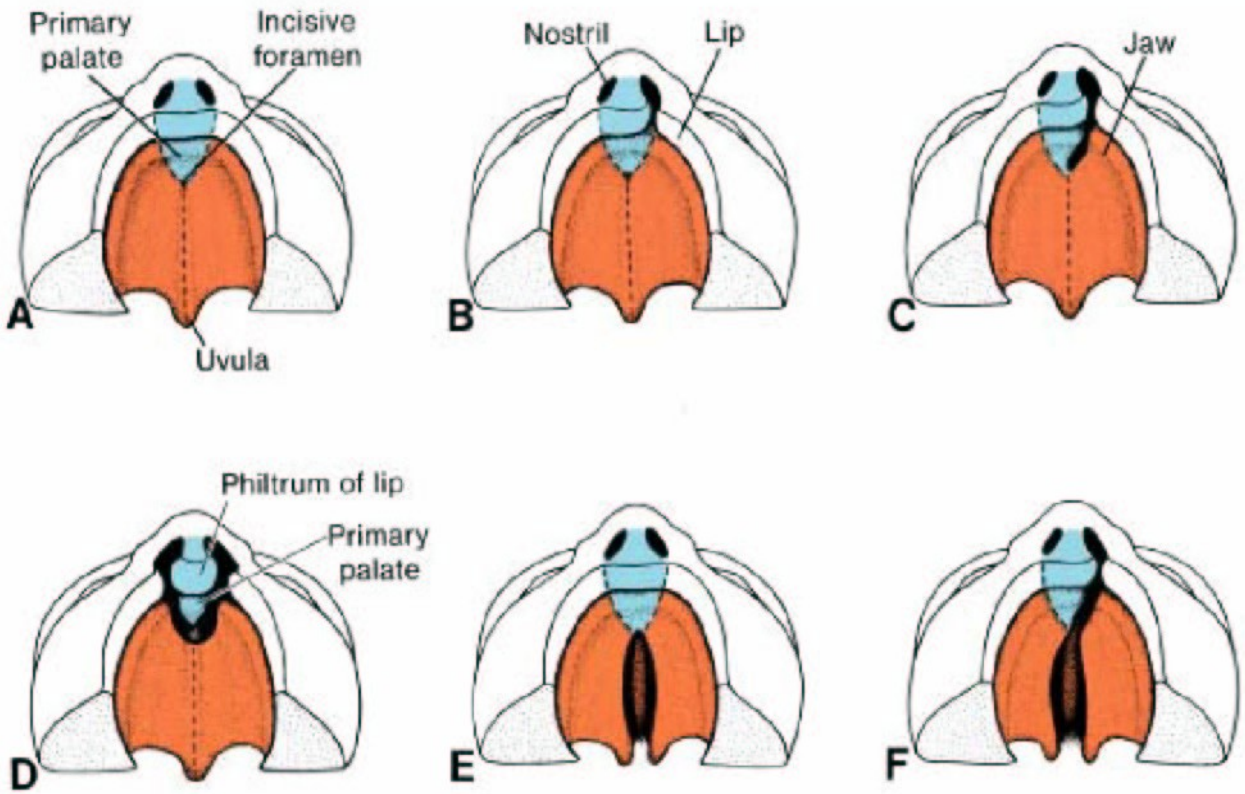
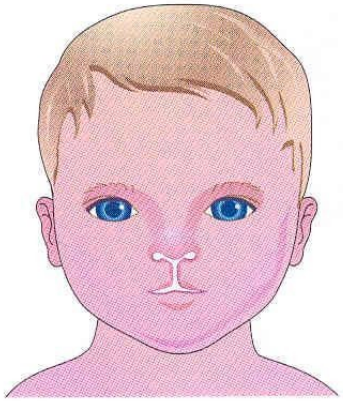
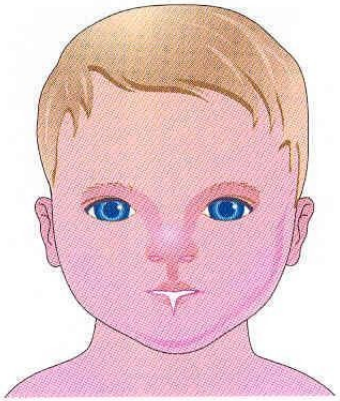


Figure 15.28 Ventral view of the palate, gum, lip, and nose. **A.** Normal. **B.** Unilateral cleft lip extending into the nose. **C.** Unilateral cleft involving the lip and jaw and extending to the incisive foramen. **D.** Bilateral cleft involving the lip and jaw. **E.** Isolated cleft palate. **F.** Cleft palate combined with unilateral anterior cleft lip.

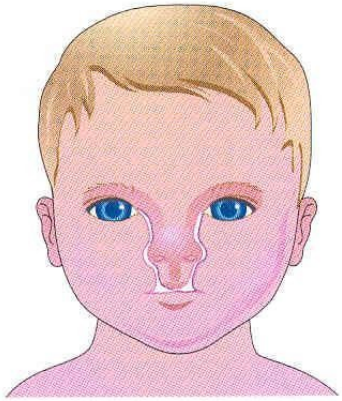
Cleft lip coupled with clefts of the anterior palate or entire palate.



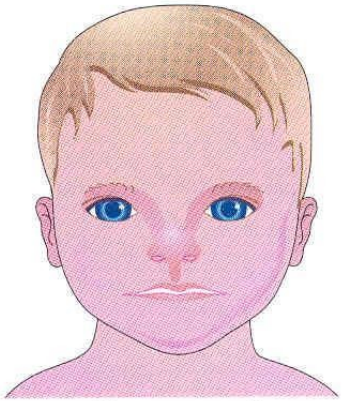
Median cleft upper lip



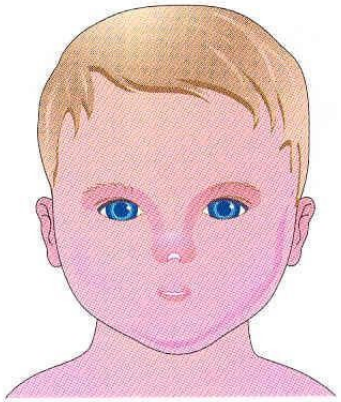
Median cleft lower lip



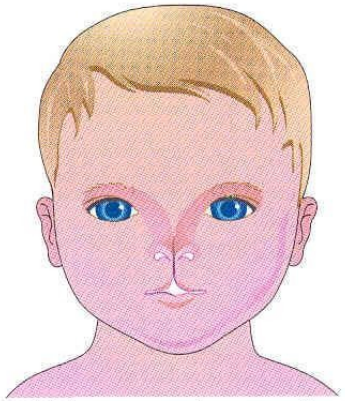
Bilateral oblique facial cleft with bilateral cleft lip



Macrostomia



Single nostril with microstomia



Bifid nose & incomplet median cleft upper lip

What matters most is how you see yourself …



Thank U
&
Good luck